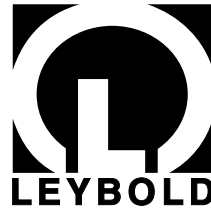


Vacuum Solutions

Application Support

Service



LEYBOLD VACUUM

GA 12.118/1.02



RGD 5/100-2

RGD 580-GE

Cold heads for GE
cryo cooler system

Leybold Part-Nos.
893 07, 891 38-1

GE Part-Nos.
2100832, 46-294100P1

Operating Instructions

Contents

Front Matter	2
Warnings.....	2
Important	3
Damage claims & return procedure.....	3
Regulatory requirements	3
1 General information	4
1.1 Introduction	4
1.2 Equipment specifications	5
2 Installation and Operation.....	8
2.1 Utility and site requirements	8
2.2 Installation Tools and Material	8
2.3 Installing the Cold Head and Compressor	8
2.4 Connecting the cold head electrical cable to the compressor.....	12
2.5 Converting the Cold Head Connection Line for an RGD 580 for Connection to an RGD 5/100-2	14
2.6 Operation	15
3 Functional checks	16
3.1 Troubleshooting list of simple problems.....	16
3.2 Checking for helium leaks	17
3.3 Checking the cold head's valve motor	18
4 Theory of operation	20
4.1 Basic principles of refrigeration	20
4.2 Cold head construction and operation.....	20
5 Spare parts	22



Service personnel shall without fail read the instructions provided for their guidance (ISO 7000-0419).

Warnings



The magnet must be ramped down before bringing ferromagnetic tools or equipment into the exam room. All tools, replacement parts, or any other objects made of iron, steel, or alloy metals could be propelled through the air by the magnetic field resulting in injury or death.



The flexlines and many of the compressor and cold head components are pressurized with helium in excess of 200 psi (14 bar); thus, before disassembling or disposing of any pressurized component, you must first vent it.

Do not install the flexlines in areas where they could be damaged or corroded, do not twist the flexline or use a flexline bending radius of < 8 inches (20 cm), do not damage the flexlines with pointed objects, and do not attempt to repair or use damaged flexlines.



The compressor has voltages which are dangerous and may be fatal if contacted. Ensure that the compressor is unplugged and switched off when making or changing electrical connections, troubleshooting, or doing repairs. To reduce the possibility of electrical shock, always connect the equipment to a low impedance ground.



Before disassembling the cold head, allow it to warm to room temperature. Removing the displacer while it is cold can result in frostbite injuries and in frost forming on the displacer.

Important:

Failure to comply with the following could cause premature failure of your cold head or compressor and void your warranty.

Failure to provide the compressor with an adequate supply of coolant can result in damage to the compressor.

Ensure that your power supply matches the compressor's power requirements.

The oil adsorber must be changed every 18,000 operating hours.

Failure to change the adsorber may result in oil contaminating the flexible gas lines and cold head. Such contamination reduces the cooling capacity and damages the cold head's displacer gaskets, control valve, and control disk.

Use only Grade 5 helium gas (99.999% pure) when adding helium to the compressor or purging the cold head and compressor. Lower grade helium contains other types of gases which freeze at cryogenic temperatures and accumulate in the cold head. This can result in deteriorating performance or failure of the cold head.

The moisture content in the helium must not exceed 10 ppm. Ensure that your plumbing is dry. We recommend using stainless steel lines since they are less likely to harbor moisture. Bake out the lines initially and keep pressured helium in the valves and lines constantly to prevent them from being contaminated with water vapor.

Don't expose the cold head to a magnetic field stronger than **300 gauss**. If the magnetic field measured at the cold-head motor is stronger than 300 gauss, the cold head may operate erratically or may even stop, and the cold-head motor could be damaged. Use suitable magnetic shielding as necessary to protect the cold-head motor from high magnetic fields.

WARNING

statements are used in this manual to prevent personal injury;

CAUTION

statements are used to prevent damage to equipment.

Damage claims & return procedure

Unpacking and Inspection

Proceed as follows to unpack and check the cold head and compressor for shipping damages as soon as you receive it.

1. Inspect the outside of each shipping container for visible damage. If you will be making a damage claim, keep the shipping container and packing materials.
2. Carefully unpack the cold head, compressor, and flexlines and inspect them for damage.
3. If you find any evidence of damage, proceed as follows:

Save the shipping container, packing material, and damaged part for inspection.

Notify the carrier that made the delivery within 7 days of delivery.

File a claim with the carrier for the damage. Any damage in transit is the responsibility of the carrier because all equipment is transported from our factory by private carriers.

Contact your service depot concerning replacement parts.

Shipping

If you ship the cold head or compressor, proceed as follows:

Reinstall the dust caps on all Aeroquip fittings to protect them from activation, damage, and contamination during shipping.

Ensure that the compressor and cold head are packed securely and mounted properly in a protective shipping crate so that they aren't subject to jarring or vibration.

Label the shipping container to caution against tipping the compressor on its side or upside down during shipping.

Regulatory requirements

UL and CSA Approbation

The RGD 5/100-2 and the RGD 580-GE cold heads and the flexlines meet the requirements of Underwriters Laboratories (UL) and the Canadian Standards Association (CSA).

Thus these appliances are able to meet the safety levels required of medical equipment.

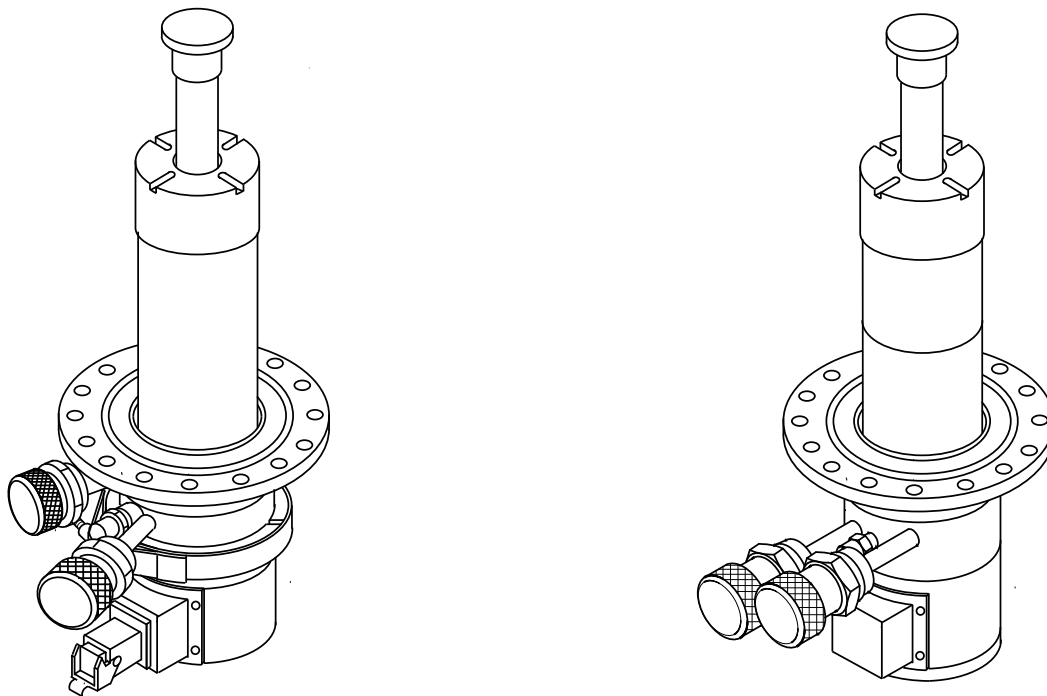


Fig.1 RGD 5/100-2 and RGD 580-GE cold heads

1 General information

1.1 Introduction

The cold head and compressor function together to produce temperature as low as 10 K.

This manual contains all of the information you need to install, operate, and maintain your cold head. It also includes a troubleshooting section, a description, and a parts list.

In addition this manual contains all information that is specific for GE cryo cooler systems.

All information you need to install, operate and maintain your compressor you will find in the compressor manuals, which are added to this manual or are delivered with the compressor.

See Section 4 for a detailed description of the cold head.

The RGD 5/100-2 and the RGD 580-GE are two-stage cold heads*. The cold head's first stage produces a temperature of between 35 and 60K and its second stage produces a temperature of between 10 and 20K.

The cold head requires a suitable compressor to operate. The compressor provides the power and the high-pressure helium needed to produce the low temperatures in the cold head. Helium gas is circulated between the cold head and compressor through flexlines.

The cold head is started and stopped on the compressor. Failures are indicated on the front of the compressor if one of the compressor's protection switches trips.

The cold head/compressor system requires very little maintenance. The oil adsorber in the compressor must be replaced every 18,000 operating hours. It is suggested to exchange the cold head displacer or the complete cold head after 9,000 operating hours in order to maintain optimum conditions for future operation.

We recommend that you keep a daily operating log for use in scheduling maintenance and identifying any degradation in performance. A blank operating log is included at the end of the manual for this purpose.

*The RGD 580-GE was delivered before June 1994, it has been replaced by the RGD 5/100-2.

1.2 Equipment Specifications

Table 1-A

Cold Head RGD 5/100-2 Technical Data

Refrigeration capacity

1st stage at 50 K > 40 W

2nd stage at 12 K > 2 W

Lowest attainable temperature

1st stage approx. < 35 K

2nd stage approx. < 10 K

Permissible ambient temperature 40-105 °F, 5-40°C

He filling pressure at room temperature 15 -16 bar
218 - 232 psig
1.5 - 1.6 MPa

Opening pressure of safety valve 28 bar
400 psig
2.8 MPa

He connections: Aeroquip screw couplings $\frac{3}{4}$ " , male

Weight excluding power cord 24.7 lbs, 11.2 kg

Standard Specification

The cold head is filled with helium. The Aeroquip screw couplings are sealed with protective caps.

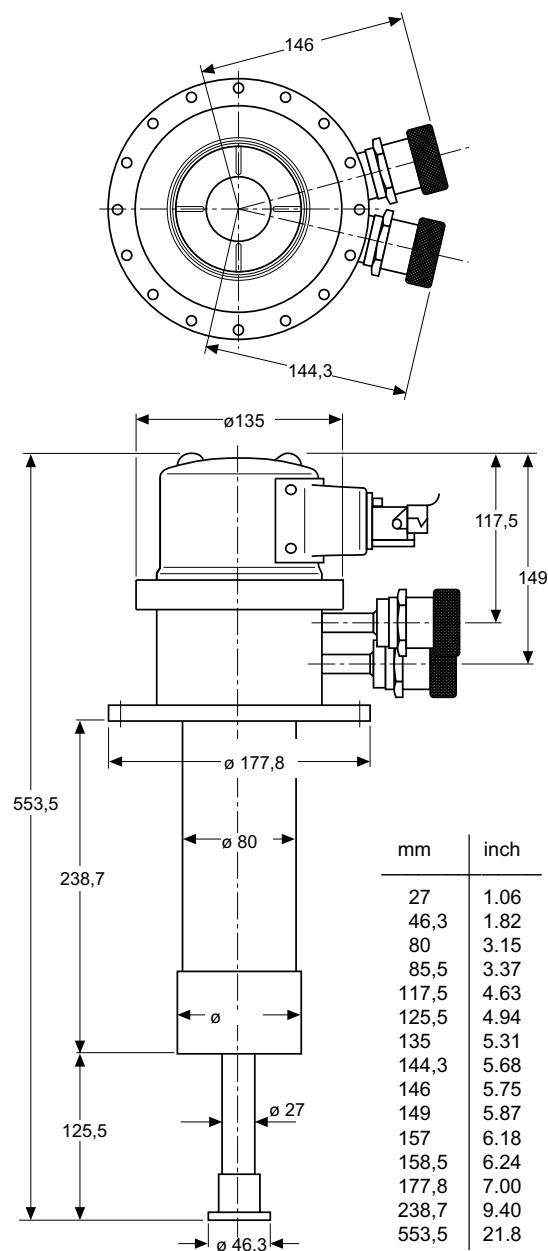


Fig. 2 RGD 5/100-2 dimensional drawing; Dimensions in mm

Table 1-B
Cold Head RGD 580-GE Technical Data

Refrigeration capacity

1st stage at 50 K > 40 W

2nd stage at 12 K > 2 W

Lowest attainable temperature

1st stage approx. < 35 K

2nd stage approx. < 10 K

Permissible ambient temperature 40-110 °F, 5-45°C

He filling pressure at room temperature 15 -16 bar
218 - 232 psig
1.5 - 1.6 MPa

Opening pressure of safety valve 28 bar
400 psig
2.8 MPa

He connections: Aeroquip screw couplings $\frac{3}{4}$ ", male

Weight excluding power cord 31 lbs, 14 kg

Standard Specification

The cold head is filled with helium. The Aeroquip screw couplings are sealed with protective caps.

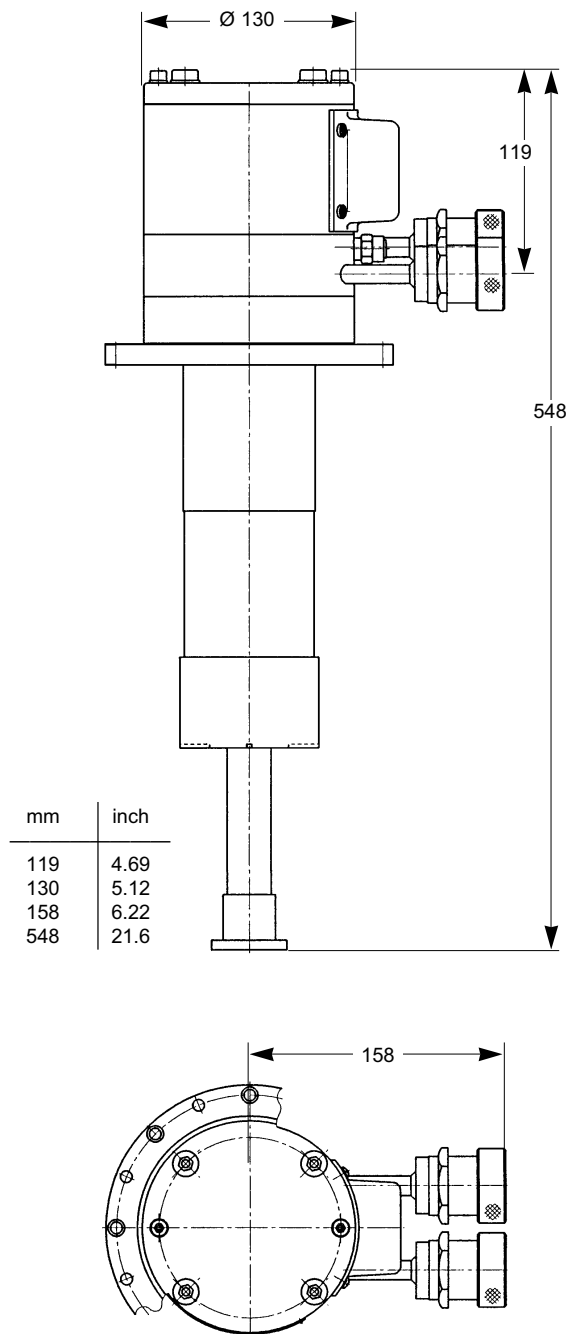


Fig. 3 RGD 580-GE dimensional drawing, Dimensions in mm

Table 1-C
Ordering Information

	Ref. No.	90° Aeroquip Adapter, 1/2-inch female to 1/2-inch male	891 73
Cold Head RGD 5/100-2	893 07		
Cold Head RGD 580-GE	891 38-1	90° Aeroquip Adapter, 3/4-inch female to 3/4-inch male	891 74
Compressor/Cold Head Cable 100 feet (two 50-foot sections with ring terminals in center)		Wrench Kit, magnetic (includes five wrenches listed below)	722-80-017
Compressor to Penetration Panel Section (50 feet)	893 72	3/4" Wrench for 1/4" Aeroquip Fitting on the Compressor	722-81-123
Cold Head to Penetration Panel Section (50 feet)	892 93	1" Wrench for 1/2" Aeroquip Fitting	722-81-018
Compressor/Cold Head Cable 40 feet (two 20-foot sections with ring terminals in center)		1 3/16" Wrench for 1/2" Aeroquip Fitting	722-81-019
Compressor to Penetration Panel Section (20 feet)	893 70	1 3/8" Wrench for 3/4" Aeroquip Fitting	722-81-020
Cold Head to Penetration Panel Section (20 feet)	892 91	1 5/8" Wrench for 3/4" Aeroquip Fitting	722-81-021
Flexlines (70-feet long, 22 meters)		Alternative wrenches (non-magnetic, stainless steel, double open-end)	
Supply Flexline	893 62	1 3/16", 19 mm	200-20-075
Return Flexline	893 63	1 3/8", 1"	200-19-190
Flexlines (20-feet long, 6 meters)		1 5/8", 1 3/16"	200-19-192
Supply Flexline	893 60		
Return Flexline	893 61		
Flexlines (100-feet long, 30.5 meters)			
Supply Flexline	893 64		
Return Flexline	893 65		

Each flexline has a 1/2" female Aeroquip screw coupling on one end for connecting to the compressor and a 3/4" female coupling on the other end for connecting to the cold head. All flexlines are filled with helium gas to 218-232 psig (15-16 bar(g)).

2 Installation and Operation

This section describes how to install the cold head and contains specific information for the installation of a GE cryo cooler system.

2.1 Utility and Site Requirements

CAUTION

Magnetic Field: Before ramping the magnet, be sure that the magnetic shield is in place if required on your particular magnet. If the magnetic field measured at the cold-head motor is stronger than 300 gauss, the cold head may operate erratically or may even stop, and its motor could be damaged. Use suitable magnetic shielding as necessary to protect the cold-head motor from high magnetic fields.

Air and Temperature Conditions

The installation site must be dust free and dry; the maximum relative air humidity is 90%.

The ambient temperature for the cold head must be between 40 and 105°F (5 and 40°C).

Helium Supply System

A helium supply system is required if you need to replace a displacer in the cold head, decontaminate the helium gas, or replace helium that has leaked out of the system. A helium supply system includes a Grade 5 (99.999% pure) helium gas bottle, a regulator, an outlet valve, and a suitable delivery line.

Flexlines and Cables

You must order the correct length flexlines and cables for your installation when you order the cold head and compressor (see Table 1-C).

Cooling Water and Power Source

See Table 2-A and compressor Operating Instructions.

2.2 Installation Tools and Material

The following tools and material are required to install the cold head, compressor, and flexlines:

Standard and metric wrench set

Wrenches for Aeroquip fittings

1" Open-end wrench

1³/₁₆" Open-end wrench

1³/₈" Open-end wrench

1⁵/₈" Open-end wrench

Clean toothbrush for cleaning the Aeroquip connectors

1/2-inch (12 mm) ID water hose

Hose clamps - 1/2" to 1¹/₈" (12 to 28 mm)

2.3 Installing the Cold Head and Compressor

The cold head can be mounted in any desired position. Mount the cold head following the GE regulations.

Warning



The magnet cooler must have a safety valve for the isolation vacuum that opens in case of

- leaks between the helium high pressure and the isolation vacuum;
- frozen gases in the isolation vacuum melt during maintenance works.

Refer to the compressor manual for the installation of the compressor and the flexlines.

The distance the compressor can be positioned away from the cold head is determined by the lengths of the flexlines and the cold head power cable.

The Aeroquip couplings on the compressor are 1/2 inch; the couplings on the cold head are 3/4 inch. Thus, each flexline must have a 1/2 inch Aeroquip coupling on one end and a 3/4 inch coupling on the other end. See Table 1-C for specifications and ordering information for the flexlines.

If you require more than 100-feet, we usually recommend using straight smooth stainless steel tubing for the return line to avoid excessive pressure drop loss.

Table 2-A Water Cooling Specifications

Equipment	Inlet temperature range	Inlet pressure	Recommended flow rate	Pressure drop new heat exchanger	Temperature rise ΔT (at 60 Hz)	Typical heat output (at 60 Hz)	Maximum heat output (at 60 Hz)
Shield Cooler Compressor RW 4000	40-80 °F 5-27 °C	Minimum 29 psi 2 bar 200 kPa	Minimum 1.0 gal/min 3.5 liter/min	4.4 psi 0.3 bar 30 kPa at minimum flow rate	39 °F 21.7 °C at minimum flow rate	16,700 BTU/h 4,900 W	18,100 BTU/h 5,300 W
		Maximum 116 psi 8 bar 800 kPa	Maximum 2.64 gal/min 10 liter/min	23.3 psi 1.6 bar 160 kPa at maximum flow rate	14 °F 7.6 °C at maximum flow rate		
Shield Cooler Compressor COOLPAK 4000	40-80 °F 5-27 °C	Minimum 29 psi 2 bar 200 kPa	Minimum 1.0 gal/min 3.5 liter/min	3 psi 0.2 bar 20 kPa at minimum flow rate	42 °F 23.3 °C at minimum flow rate	16,700 BTU/h 4,900 W	19,500 BTU/h 5,700 W
		Maximum 116 psi 8 bar 800 kPa	Maximum 2.64 gal/min 10 liter/min	9 psi 0.6 bar 60 kPa at maximum flow rate	14.7 °F 8.2 °C at maximum flow rate		

Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.

Pressure drop is measured between coolant inlet and outlet at compressor.

Temperature rise, typical and maximum heat output are reduced by approx. 15 % at 50 Hz operation.

Condensation will occur if inlet temperature is below dew point temperature; see compressor manual for details.

To set the flow rate the water throughput must be throttled in the outlet.

Check the system helium pressure after installation of the cold head and compressor. Refer to the compressor Operating Instructions. Normally you must release helium from the COOLPAK compressor unit after installation.

For the compressor installation go through Sections 2.1 to 2.6 of the compressor manual; for the electrical connection refer to Section 2.5 of the compressor manual **and** Section 2.4 of this manual.

For the RW 4000, go through Section 2.7 „Start-up“ of the compressor manual to finish the installation.

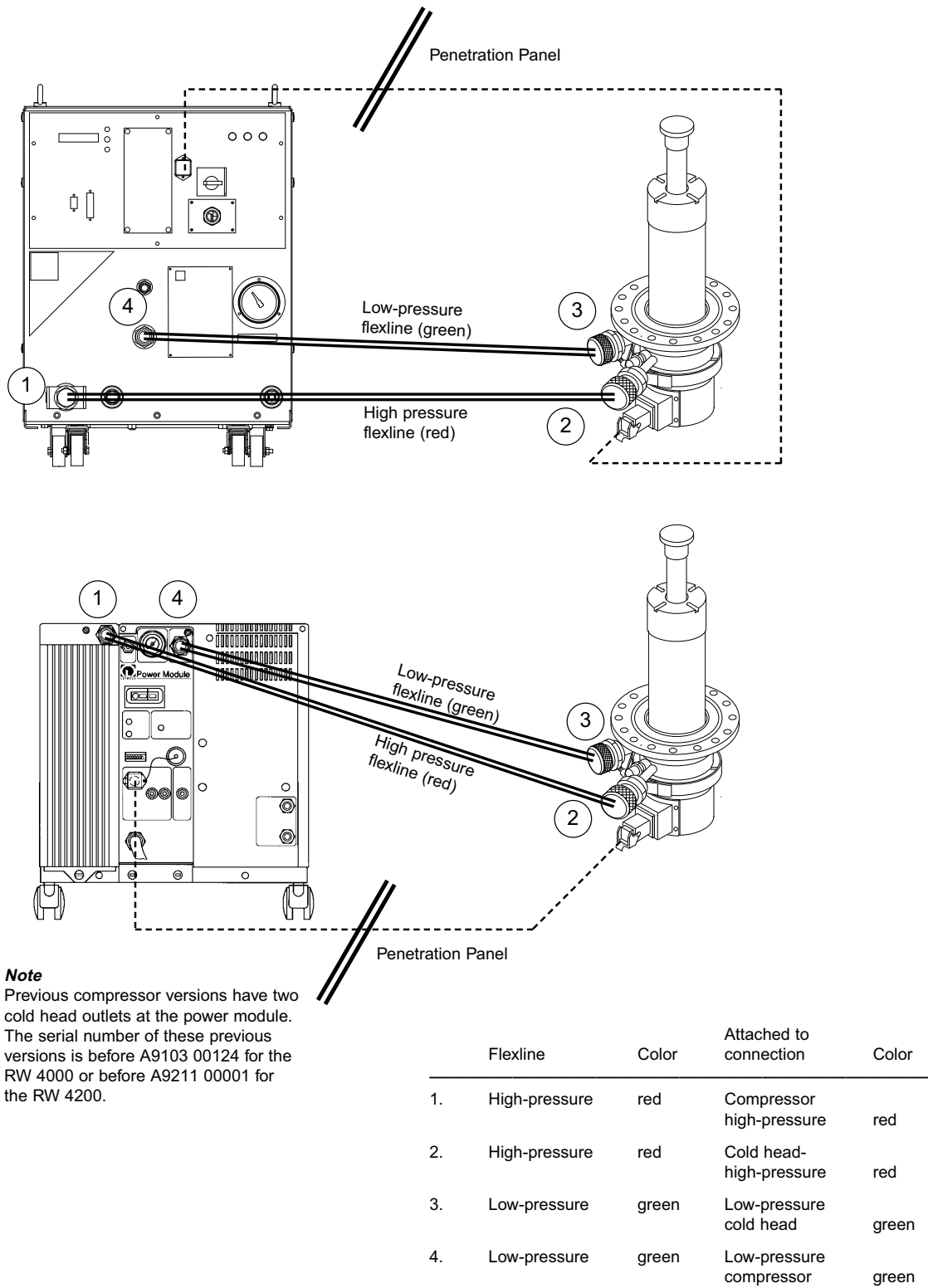
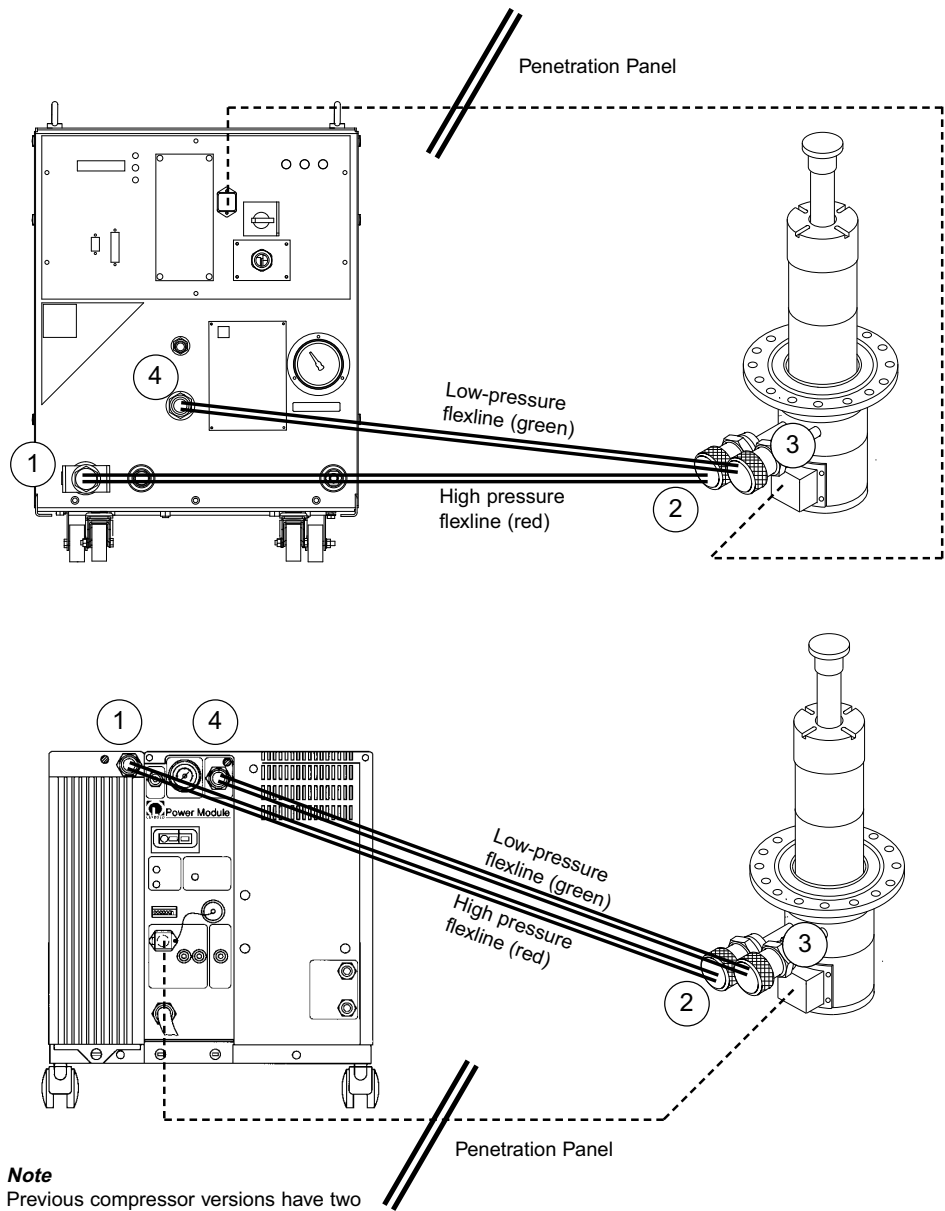


Fig. 4 Connecting the RGD 5/100-2 cold head and compressor unit



Note
Previous compressor versions have two cold head outlets at the power module. The serial number of these previous versions is before A9103 00124 for the RW 4000 or before A9211 00001 for the RW 4200.

	Flexline	Color	Attached to connection	Color
1.	High-pressure	red	Compressor high-pressure	red
2.	High-pressure	red	Cold head-high-pressure	red
3.	Low-pressure	green	Low-pressure cold head	green
4.	Low-pressure	green	Low-pressure compressor	green

Fig. 5 Connecting the RGD 580-GE cold head and compressor unit

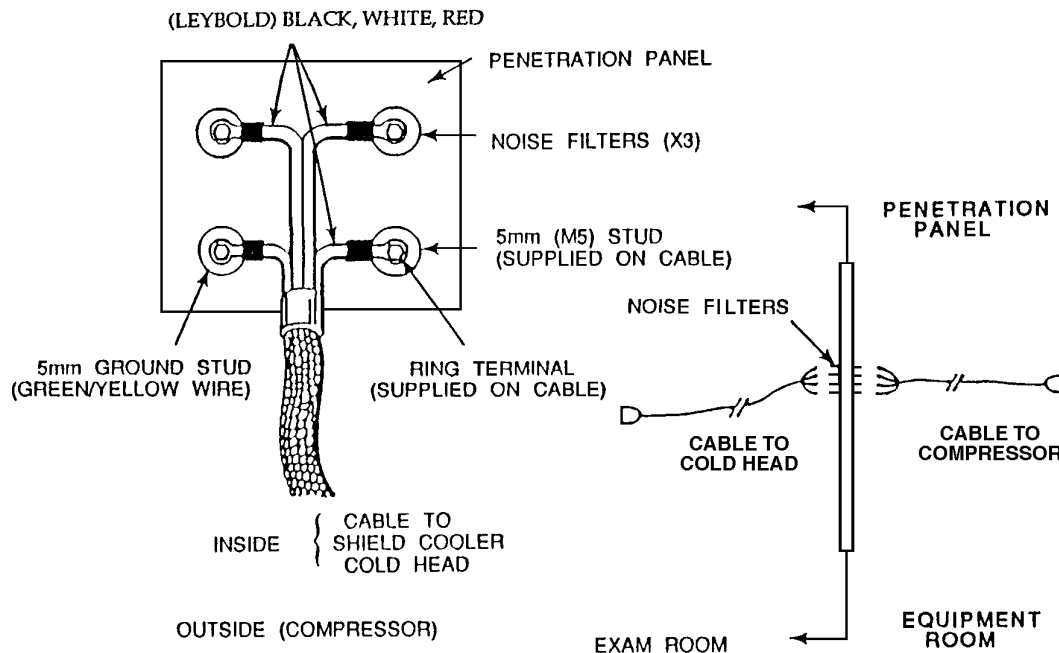


Fig. 6 Electrical cable connections

2.4 Connecting the Cold Head Electrical Cable to the Compressor

The cable is in two sections for most installations. One section (RUN #624) runs from the cold head to the penetration-panel noise filters (see Figure 6). The second section (RUN #623) runs from the other side of the penetration panel to the compressor.

Install the cable sections as follows (see Figure 6):

1. Connect the four ring terminals on the end of the cold-head electrical cable (RUN # 624) to the four studs of the noise filter on the inside of the penetration panel.

RGD 5/100-2 only

Plug the other end of this cable into the socket on the motor housing of the cold head.

RGD 580-GE only

Plug the other end of this cable onto the mating pins on the cold head ensuring that the arrow on the cluster connector is pointing toward the cold head's Aeroquip fittings and cold stages (see Fig. 7). Use the ground screw to connect the ground wire. Connect the cover to the cold head using the four screws.

2. Connect the four ring terminals on the end of the compressor electrical cable (RUN # 623) to the four studs of the noise filter on the outside of the penetration panel (equipment room side). **Make sure that the wire code is properly matched (black 1 to black 1, white 2 to white 2, red 3 to red 3, and green/yellow to green/yellow) from the inside to the outside of the penetration panel.**

Plug the other end of this cable into the COLD HEAD outlet on the compressor.

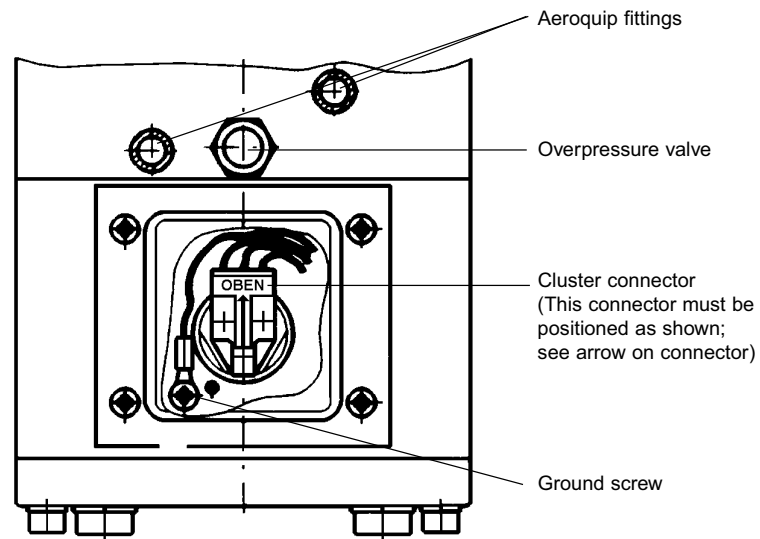
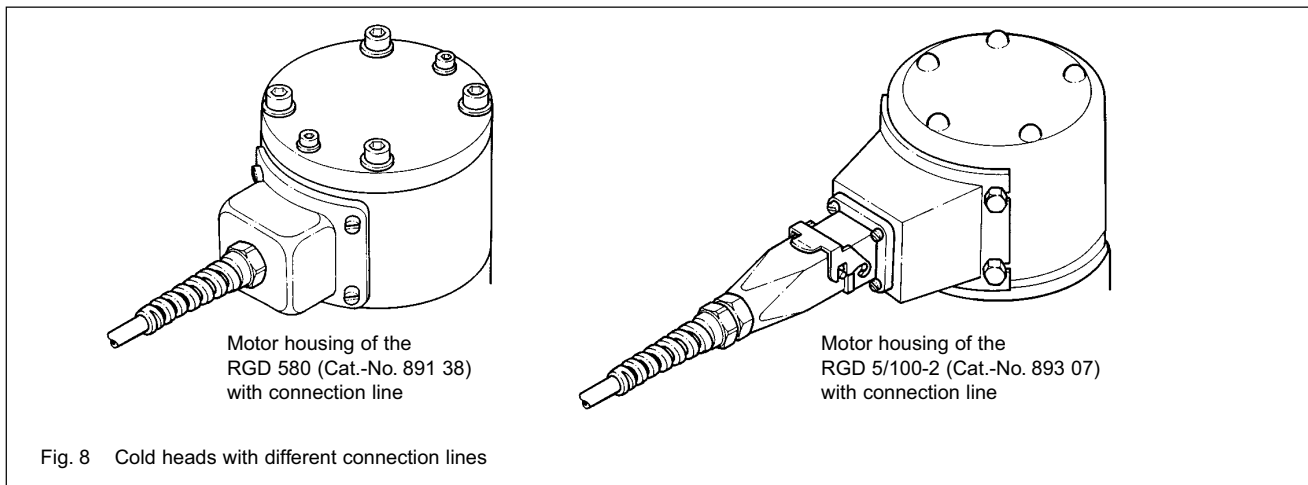


Fig. 7 Connecting the cable onto the RGD 580-GE cold head



2.5 Converting the Cold-Head Connection Line for an RGD 580 for Connection to an RGD 5/100-2

The electrical cold-head connection line is permanently fixed on the RGD 580; on the RGD 5/100-2, it is secured by a plug. Consequently, when replacing the RGD 580 with the RGD 5/100-2 either the connection line must be exchanged or a plug (HTS) must be attached to the connection line. With a permanent fixed connection line it is easier to fit the plug.

The instructions in this Section describe the fitting of a plug (HTS) to the cold-head connecting line of the RGD 580.

Order number of the conversion kit: 200 20 836.

Tools required:

3 mm and 6 mm (1/8" and 1/4") screwdrivers

Engineer's pliers, wire strippers

SW 22 (7/8") spanner

Prior to the conversion check the conversion kit and the cold head plug on the RGD 5/100-2. The conversion kit's plug must have **sockets**, because the connection line may be under voltage. The cold head's plug must have **pins**. If you find deviations please inform Leybold.

Warning



Before carrying out the conversion, switch off the compressor. Withdraw the mains plug for the compressor or withdraw the cold-head connection line on the compressor.

The conversion may only be carried out by a qualified electrician in accordance with the applicable regulations.

To carry out the conversion, the cold-head connection line must first be removed from the RGD 580.

Release the threaded conduit joint on the connection line. Release the plastic cap over the electrical connection to the cold head. Cut the four cores of the cable as close to the cold head as possible. Pull the plastic cap off the connection line and re-secure to the cold head. Leave the threaded conduit joint on the connection line, as this is required for assembly.

To fit the plug (HTS), see Fig. 9.

Strip the insulation from the connection line. 40 mm (1 9/16") of outer insulation must be removed, and 7 mm (9/32") from the ends of the cores.

Unscrew the plug union. It will no longer be required.

Unscrew the screw and dismount the plug.

Remove the inner ring of the rubber ring to obtain the correct inside diameter.

Push the washer, rubber ring, washer and connector housing onto the connection line.

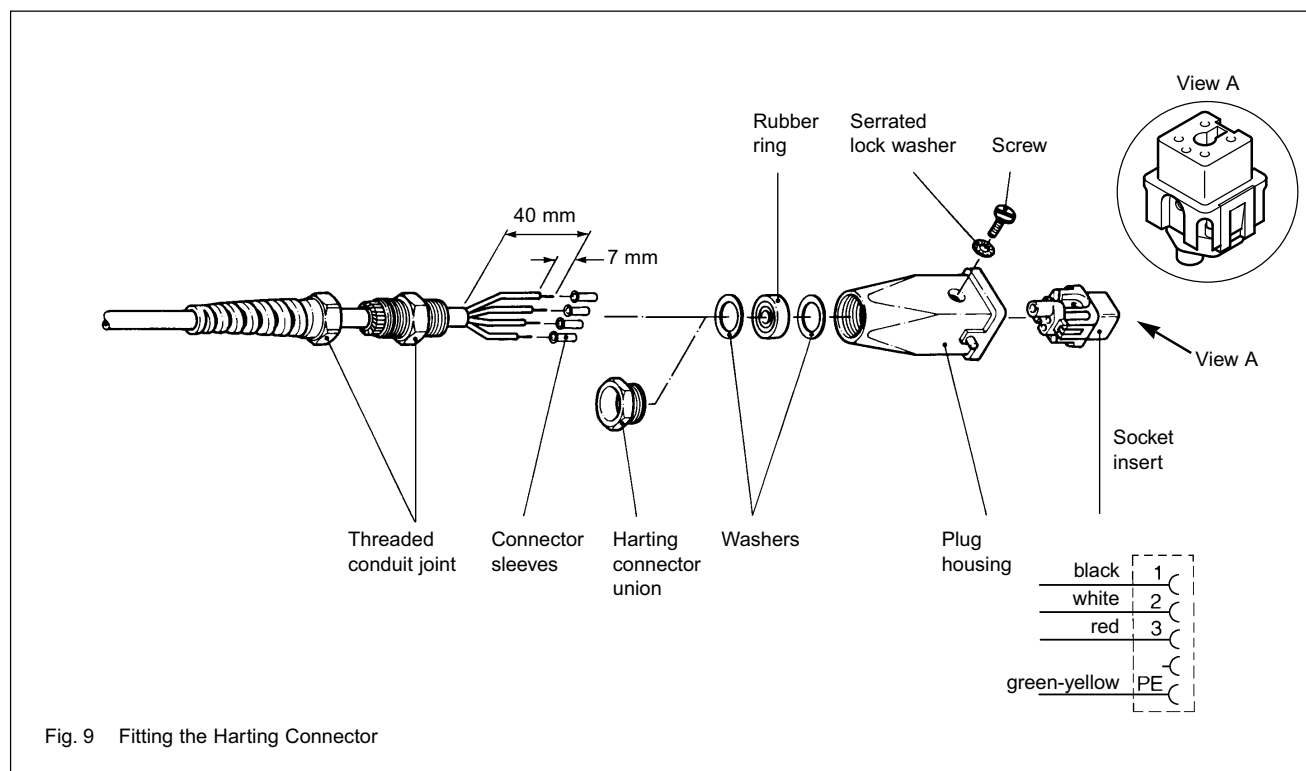
Push the connector sleeves over the ends of the cores, press them together, and connect the ends of the cores to the socket insert as follows:

Green-yellow to PE

Black to 1

White to 2

Red to 3.



Push the plug housing over the socket insert, and secure using the screw and serrated lock washer.

Push the washer, rubber ring and washer into the plug housing.

Tighten the threaded conduit joint.

The cold-head connection line is now ready for connection to the RGD 5/100-2.

2.6 Operation

The cryo cooler system is operated at the compressor.

Cooldown Time

The total cooldown time of the cold head from 300K (room temperature) to 10K depends on the refrigerating capacity of the cold head, the thermal radiation load, and the mass attached to the cold head.

3 Functional Checks

The system operates properly if

- the helium filling pressure is correct at standstill of the system (see compressor Operating Instructions),
- the operating pressure is between the specified values (see compressor Operating Instructions), and if
- no FAIL. LEDs are lit or SYSTEM ERROR is displayed at the compressor.

Further checks are not necessary.

WARNING



The flexlines and many of the compressor and cold head components are pressurized with helium in excess of 200 psi (14 bar); thus, before disassembling or disposing of any pressurized component, you must first vent it. Don't install the flexlines in areas where they could become damaged or corroded, don't twist the flexline or use a flexline bending radius of <8 inches (200 mm), don't damage the flexlines with pointed objects, and don't attempt to repair or use damaged flexlines. See the other important warnings on the inside front cover of this manual before working on the cold head or compressor.

IMPORTANT

Cold head repairs shouldn't be performed when the cold head is installed to a system. It is suggested to exchange the cold head displacer or the complete cold head after 9,000 operating hours in order to maintain optimum conditions for future operation. All maintenance work has to be done by specially trained persons.

3.1 Troubleshooting list of Simple Problems

When you have a problem with the cold head/compressor, we recommend that you first go through the following checklist of simple problems before assuming that source of the problem is a cold head or compressor failure.

Is the cold head receiving power?

Ensure that the compressor is connected to the correct source of AC power and that the transformer setting matches your power source (refer to compressor Operating Instructions).

Ensure that your AC power source is OK.

Ensure that the cable connecting the cold head to the compressor is plugged in securely and that the wiring at the penetration panel is correct.

Is the compressor rotating in the correct direction?

Has the displacer been replaced in the past 9,000 hours?

Are the cold head and compressor being properly cooled?

Is your shield temperature monitoring equipment operating properly?

Is the helium pressure adequate? Check for helium leaks.

Is the helium contaminated?

Use only Grade 5 helium gas (99.999% pure) when adding helium to the compressor, and ensure that your helium delivery system (lines, regulator, etc) doesn't contain moisture or air.

Change the oil adsorber every 18,000 hours (see compressor Operating Instructions).

Are the flexlines installed correctly so that one line connects the cold head's high-pressure port to the compressor's high-pressure port and the other line connects the low pressure ports?

Is the valve motor on the cold head operating properly (see Section 3.3)

3.2 Checking for Helium Leaks in the Compressor and Cold Head

If the helium circuit is leaking, the pressure shown on the compressor's pressure gauge will be low and the cold head won't reach its normal operating temperature. If the helium pressure is very low, the compressor shuts down.

Check the helium pressure when the compressor is shut-down and all components are at room temperature.

The best way to leak check is with a mass spectrometer leak detector with a sniffing probe. If a leak detector isn't available, you can check for leaks using a soap solution or visually inspect all the joints in the compressor for cracks.

Proceed as follows to isolate helium leaks:

1. While the flexlines are connected to the cold head and compressor, check for leaks at the self-sealing Aeroquip gas couplings on the cold head and then at the couplings on the compressor.

2. Now isolate each system subassembly by disconnecting the self-sealing gas couplings at both the cold head and compressor, and leak checking each system subassembly. Be sure to check each joint including the pressure safety valve and the low-pressure switch on the compressor, and any tubing that has been rubbing during operation. Don't assume that a single leak is the source of the problem.

Helium leak detectors can be misleading. Leaking gas can form patterns that indicate leaks at sound joints. Also, heavy leakage or a high gas concentration can make it difficult to isolate the leak.

If a mass spectrometer leak detector isn't available, use a commercial soap solution specially prepared for leak testing. We recommend Snoop LC solution. Such solutions leave little residue and are manufactured under controls to enhance their effectiveness; however, soap-solution testing may miss small leaks.

Proceed as follows when testing for leaks with a soap solution:

1. Fully coat the joint being tested.
2. Allow time for bubbles to form at small leaks.
3. Look carefully for the smallest bubble formations.

CAUTION

After soap-solutions testing, use water to wash all soap residue from joints and fittings. Residue in the poppet face of an open Aeroquip fitting can contaminate the displacer.

After eliminating the leak, add helium to the compressor as described in the compressor Operating Instructions.

3.3 Checking the Cold Head's Valve Motor

If the cold head motor is malfunctioning, you may notice a gradual warming of the cold head. Also the audible operating cycle will be slower than 2.4 Hz at 60 Hz mains frequency.

Checking the cold head's valve motor is **not** a GE field procedure. In case of a malfunction the cold head must be removed from the magnet and sent to an authorized repair depot to have it checked and repaired.

1. Using an ohmmeter, check the valve motor resistance as follows:

- a. Shut off power to the compressor and disconnect it from its AC power source.
- b. Unplug the power cord connecting the cold head to the compressor.
- c. Remove the cover from the cold head's junction box and unplug the cluster connector.
- d. Set the ohmmeter to a suitable scale and measure the resistances between the pin combinations listed below. The pins are located on the cold head as shown in Figure 10. If the meter reading isn't within the resistance tolerance listed below, replace the drive assembly.

Cold Head Pins	Resistance for RGD 5/100-2	Resistance for RGD 580-GE
1-2	630 Ω + 10%, - 5%	700 Ω $\pm$ 10%
2-3	630 Ω + 10%, - 5%	700 Ω $\pm$ 10%
1-3	630 Ω + 10%, - 5%	1400 Ω $\pm$ 10%

e. Hold one ohmmeter lead to the valve-motor housing ground connection and the second to pin 1. If the meter indicates continuity, check the wire connections for short circuits; otherwise, replace the cold head.

f. Repeat Step 1e for pins 2 and 3.

2. If the cable between the cold head and compressor isn't connected properly, the valve motor will either rotate in the wrong direction or it won't rotate at all. If the motor is rotating in the wrong direction, the cold head won't reach its minimum operating temperature or its specified capacity. If the valve motor isn't rotating at all, you won't hear the gas compression and the cold head won't cool down.

To check the valve motor rotation you need the motor power supply, Leybold P/N 200 19 619 (200/240 V, 50/60 Hz, 1~).

Check the valve motor rotation as follows:

a. Let the cold head warm up and disconnect the flexlines at the cold head.

Warning



The cold head is pressurized with helium. Vent the cold head before disassembling it. To do so, screw an Aeroquip-swagelock adapter into one of the Aeroquip screw couplings.

b. Remove the motor from the cold head.

c. Set the switch of the motor power supply for the appropriate coldhead motor type (Position 2.0 μ F for RGD 5/100 or 820 Ω / 0.8 μ F for RGD 580).

d. Connect the cold head power cord to the motor power supply and switch on the motor power supply.

Caution

Don't run the compressor when the flexlines aren't connected. Running without flexlines could result in oil contaminating the helium or overloading the compressor.

e. Observe the rotation of the control valve; when facing the sealing surface of the control valve, the valve should rotate clockwise as shown in Figure 10.

If the control valve rotated the wrong direction, it means that the wire color codes aren't matched between the inside and outside of your penetration panel.

Remount the cold head.

After having checked the valve motor you must purge the cold head with helium.

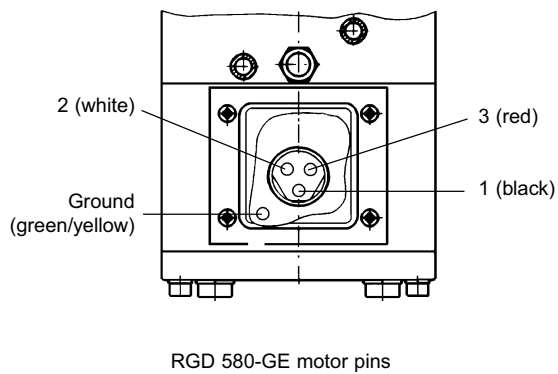
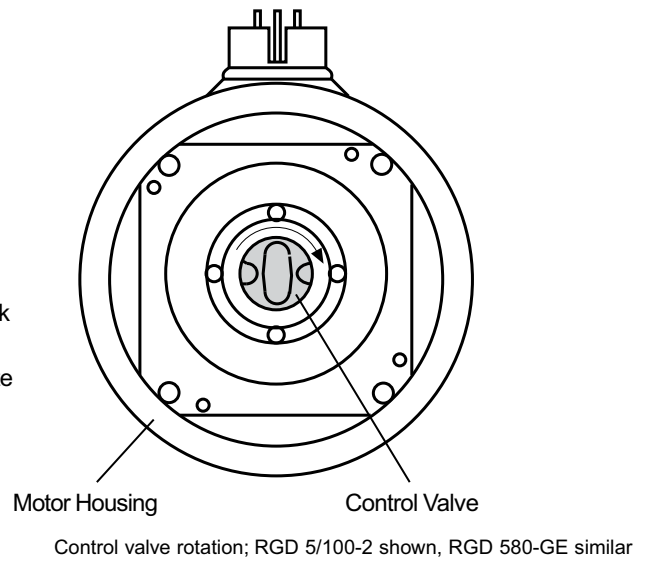
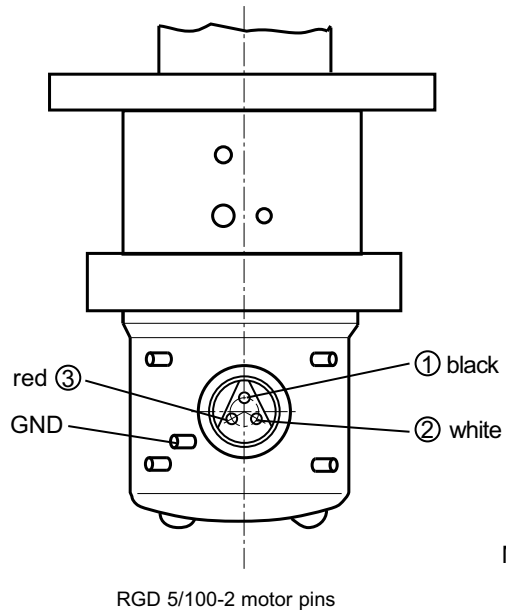


Fig. 10 Checking the cold head

4 Theory of Operation

The RGD 5/100-2 and the RGD 580-GE are two-stage cold heads (see Figure 11). Their first stage produces a typical operating temperature of between 35 and 60K and their second stage produces a temperature of between 10 and 20K. They require a compressor to provide the power and the high-pressure helium needed to produce the low temperatures in the cold head.

4.1 Basic Principles Of Refrigeration

The cold head is similar in operation to the refrigerator in your home. In a home refrigerator, gas is compressed, the heat of compression is removed by an air-cooled heat exchanger, and the gas is then allowed to expand. The expanding gas removes heat energy from the expansion cylinder, thus cooling the expansion cylinder below room temperature. The gas is then returned to the compressor and the cycle is repeated.

This simple compression-expansion process is also used in the cold head's refrigeration system. However, helium is used to produce extreme low temperatures because helium remains gaseous at temperatures down to 4.2K (-452°F). Helium requires a more complex mechanical expansion device.

The Leybold cryo-refrigeration system is based on the Gifford-McMahon principle using a closed helium gas cycle. In this type of system, the expansion cylinder (cold head) is separated from the compressor and connected via flexible gas lines. The greatest advantage of this system is that the cold head can be mounted in any desired mounting position, independent of its compressor.

4.2 Cold Head Construction and Operation

Figure 11 shows the internal components of a typical cold head. The flow of low- and high-pressure helium through the cold head is controlled by the control valve. The control valve is rotated clockwise at 72 rpm at 60 Hz mains frequency by an electric synchronous stepping motor. As the control valve rotates on top of the control disk, holes in the valve align with different holes in the disk. When one set of holes are aligned, high-pressure helium flows into the expansion chamber. As the valve rotates, this high-pressure channel is sealed and another set of holes align. This second set of aligned holes forms a channel which allows the helium in the expansion chamber to expand, cool, and escape.

The movement of the displacer and hence the volume of the expansion chamber is also controlled by the control valve. As the control valve rotates, it channels high and then low pressure helium into a control chamber. This alternating high and low helium pressure in the control chamber forces the control piston and attached displacer to move up and down at a frequency of 2.4 Hz at 60 Hz mains frequency.

The helium connections have self-sealing couplings which keep the cold head under pressure when the gas lines are removed. A relief valve relieves pressure from the cold head if it is exposed to extremely hot ambient temperatures when shut off. This relief valve also protects the cold head from internal damage if the flexlines are removed while the helium inside the cold head is expanding.

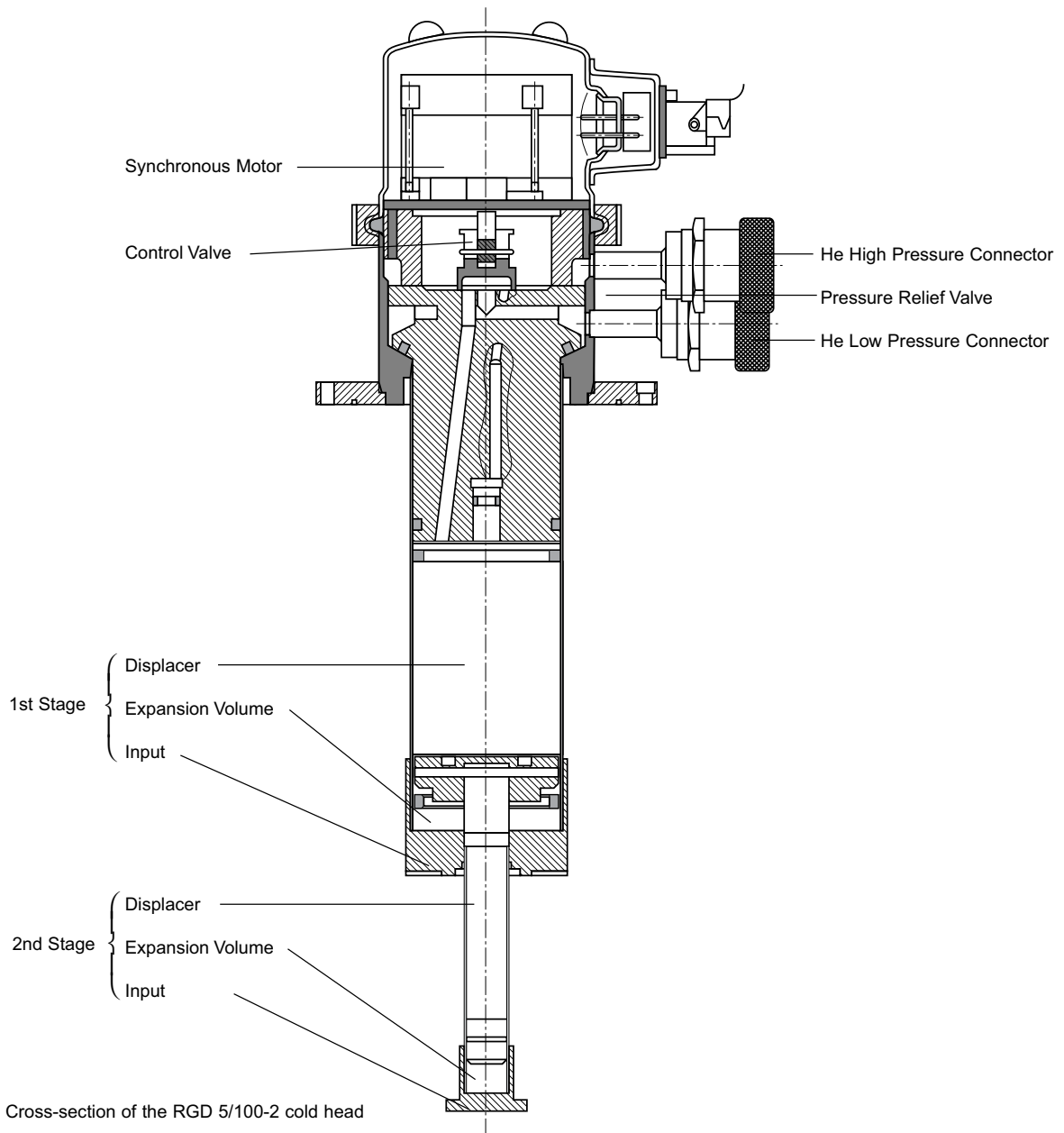


Fig. 11 Cross-section of the RGD 5/100-2 cold head

5 Spare parts

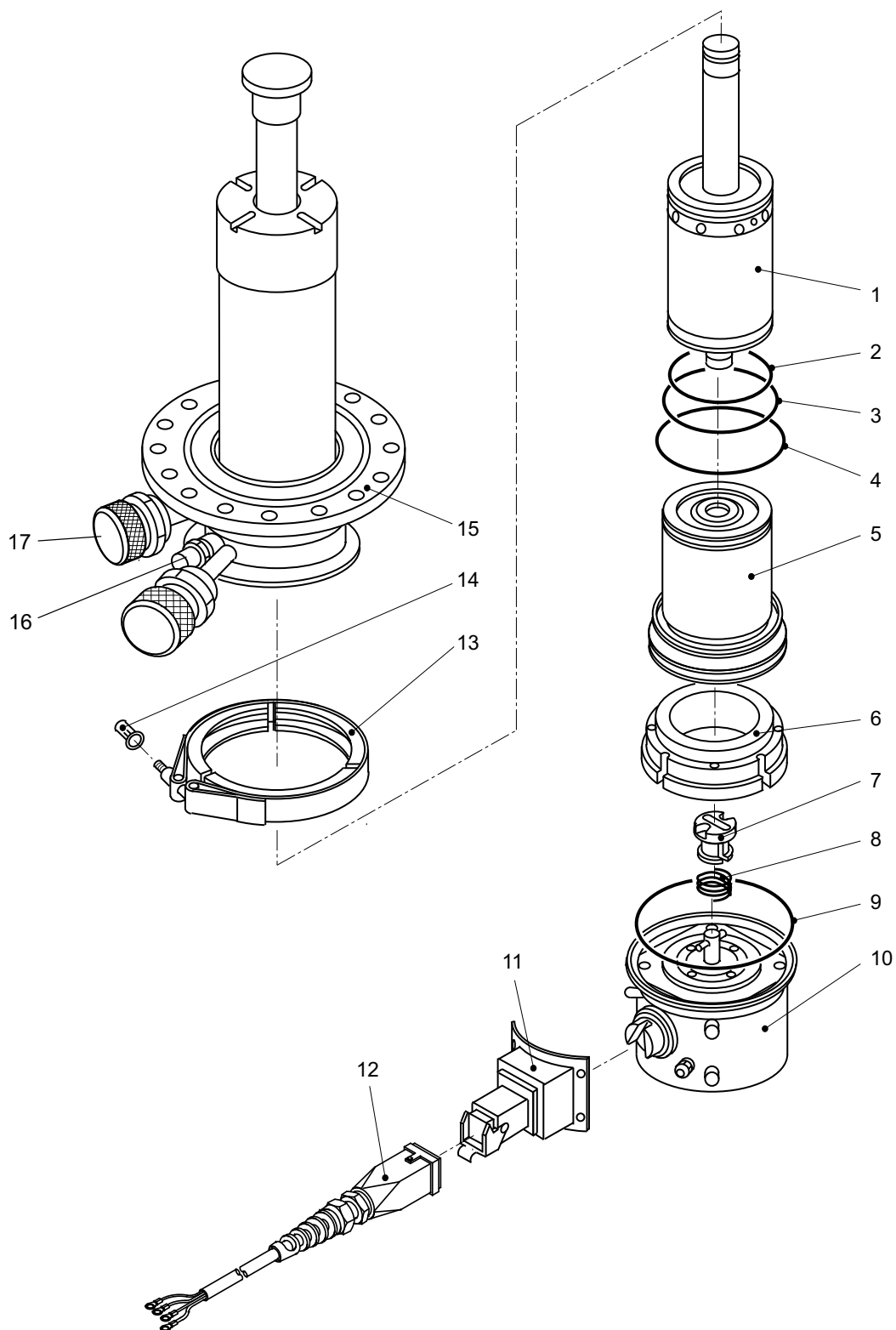


Fig. A RGD 5/100-2

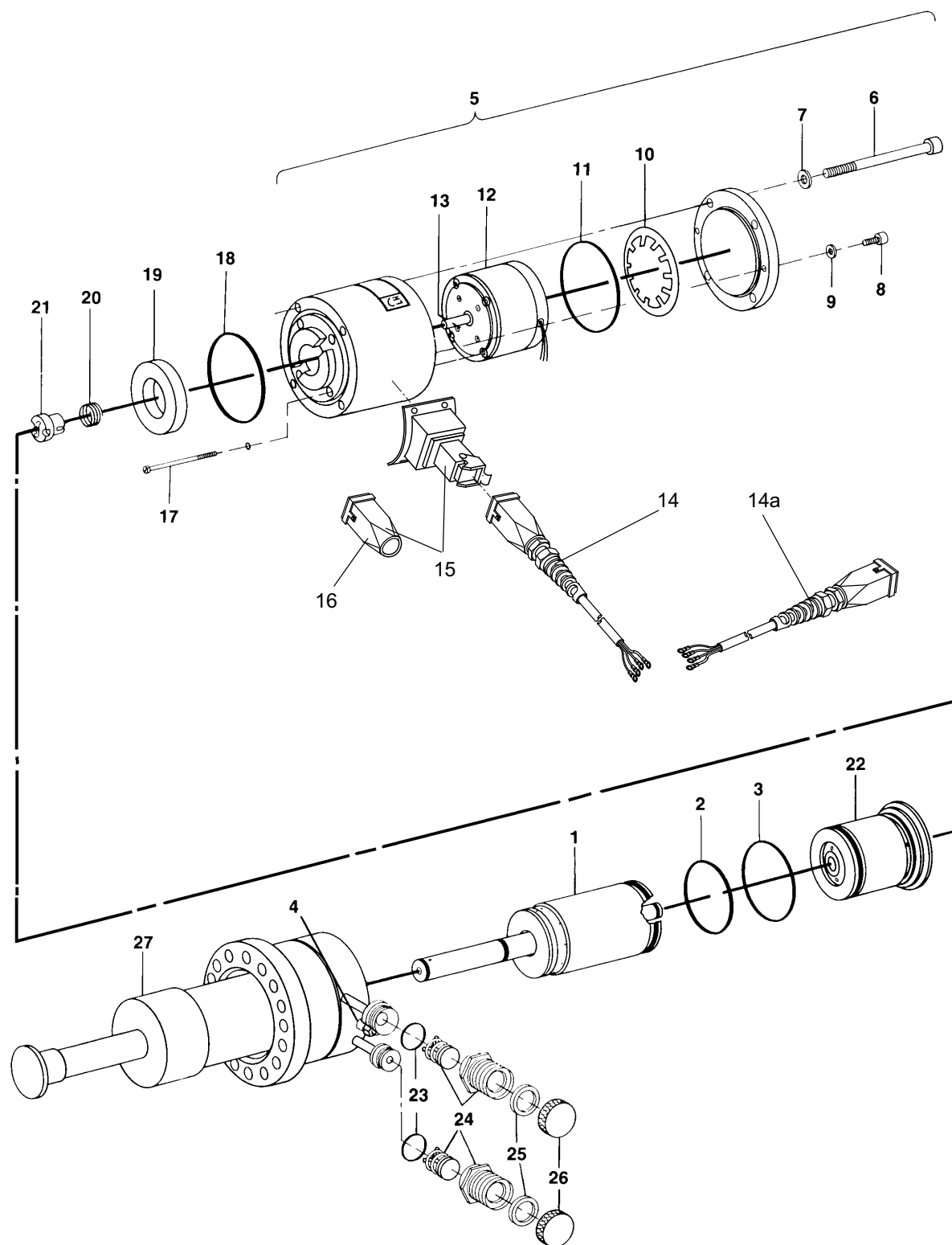


Fig. B RGD 580-GE

EEC Manufacturer's Declaration

in the sense of EEC Directive on Machinery 89/392/EWG, Annex IIb



We - LEYBOLD Vacuum GmbH - herewith declare that operation of the incomplete machine defined below, is not permissible until it has been determined that the machine into which this incomplete machine is to be installed, meets the regulations of the EEC Directive on Machinery.

At the same time we herewith certify conformity with EEC Directive on Low-Voltages 73/23/EWG.

As regards the EMC Guidelines, see the relevant remarks in the Conformance Declaration for the compressor unit.

Designation: Cold Head

Models:

RGD 5/100-2, RGD 580-GE

Catalogue numbers:

893 07, 891 38-1

Applied harmonized standards:

- EN 292-1 u. -2: 1991
- EN 1012-2: 1996
- EN 60204-1: 1992

Applied national standards and technical specifications:

- VBG 4/04.79 Elektr. Betriebsm. u. Anlagen
- DIN VDE 0113-1/02.86 Elektr. Ausrüstung
- AD-Merkblatt A2, B0, W0

Cologne, Feb. 14, 1997

Dr. Mattern-Klosson, Business Area Manager
Cryo Technology

Cologne, Feb. 14, 1997

Veit, Designer
Cryo Technology

LV/GK.0184.02.02.97

Operating Log

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